

Finite Element Method

Divergence Theorem
MECH4103 — Spring 2026

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- In the following lectures, we will retrace the same path that we have just traversed for one-dimensional problems for multidimensional problems.
- We will again follow the same roadmap, starting with the development of the strong form and weak form in this lecture.
- We will now consider a narrower class of problems called **scalar problems** because the unknowns are scalars like temperature or a potential.
- The methods that will be developed in these chapters apply to problems such as steady-state heat conduction, ideal fluid flow and electric fields.
- One complication in extending the methods to multidimensions lies in notation. In multidimensions, variables such as heat flux and displacement are vectors.
- We first review some important notations for 2D problems.

Then a vector $\vec{q}$ can be expressed in terms of its components by

$$\vec{q} = q_x \vec{i} + q_y \vec{j}, \quad (1)$$

where q_x and q_y are the x and y components of the vector, respectively. When we get to the derivation of finite element equations, it becomes convenient to use matrix notation. A column matrix can be used to describe a vector $\vec{q}$ by listing the components of the vector in the order as shown below:

$$\mathbf{q} = \begin{bmatrix} q_x \\ q_y \end{bmatrix}. \quad (2)$$

- Scalar product

$$\vec{q} \cdot \vec{r} = q_x r_x + q_y r_y. \quad (3)$$

- Or in matrix form

$$\mathbf{q}^T \mathbf{r} = \begin{bmatrix} q_x & q_y \end{bmatrix} \begin{bmatrix} r_x \\ r_y \end{bmatrix} = q_x r_x + q_y r_y. \quad (4)$$

- So writing the scalar product in terms of the matrices requires taking the transpose of the first matrix.

- Another important operation in vector methods is the **gradient**. The gradient provides a measure of the slope of a field, so it is the two-dimensional counterpart of a derivative. The gradient vector operator is defined by

$$\vec{\nabla} = \underbrace{\left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} \right)}_{\vec{\nabla}}. \quad (5)$$

The gradient of the function $\theta(x, y)$ is obtained by applying the gradient operator to the function, which gives

$$\vec{\nabla} \theta = \vec{i} \frac{\partial \theta}{\partial x} + \vec{j} \frac{\partial \theta}{\partial y}. \quad (6)$$

- The scalar product of the gradient operator with a vector field gives the **divergence** of the vector field. The term divergence probably originated in fluid mechanics, where it refers to the flow leaving a point. The divergence of a vector is obtained by

$$\vec{\nabla} \cdot \vec{q} = \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} \right) \cdot (q_x \vec{i} + q_y \vec{j}) = \frac{\partial q_x}{\partial x} + \frac{\partial q_y}{\partial y} \equiv \text{div } \vec{q}. \quad (7)$$

- Notice that the divergence of a vector field is a scalar.

- The matrix form of the divergence is written by replacing the dot in the scalar product by a transpose operation, so

$$\operatorname{div} \vec{q} = \vec{\Delta} \cdot \vec{q} = \nabla^T \mathbf{q}. \quad (8)$$

- The gradient operator is defined as a column matrix. So

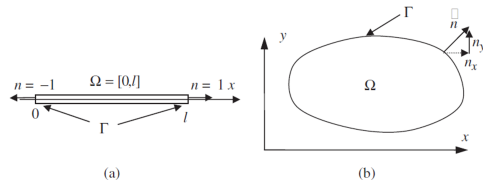
$$\nabla = \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{bmatrix} \quad \text{and} \quad \nabla \theta = \begin{bmatrix} \frac{\partial \theta}{\partial x} \\ \frac{\partial \theta}{\partial y} \end{bmatrix}. \quad (9)$$

- It is important to notice that when we write the gradient operator in vector notation, an arrow is placed on the inverted del; in matrix notation, the arrow is omitted.

- The two-dimensional equations will be developed for a body of arbitrary shape.
- The boundary of the domain is denoted by Γ . Notice that our nomenclature is identical to that in the previous lectures, but now the symbols refer to more complicated objects.
- The unit normal vector to the domain, denoted by $\vec{n}$, is given by

$$\vec{n} = n_x \vec{i} + n_y \vec{j},$$

- n_x and n_y are the x and y components of the unit normal vector, respectively; this vector is also called the **normal vector** or just the **normal**.



(a) One-dimensional domain and (b) two-dimensional domain.

(a) One-dimensional domain and (b) two-dimensional domain.

- For any C^0 integrable function in a one-dimensional domain Ω with boundaries Γ we have

$$\int_{\Omega} \frac{d\theta(x)}{dx} dx = (\theta n)|_{\Gamma}. \quad (10)$$

- The generalization of this statement to multidimensions is given by **Green's theorem**, which states:

If $\theta(x, y) \in C^0$ and integrable, then

$$\int_{\Omega} \vec{\nabla} \theta d\Omega = \oint_{\Gamma} \theta \vec{n} d\Gamma \quad \text{or} \quad \int_{\Omega} \nabla \theta d\Omega = \oint_{\Gamma} \theta \mathbf{n} d\Gamma. \quad (11)$$

Using the above, we will now develop a theorem that relates the area integral of the divergence of a vector field to the contour integral of a vector field, which is called the *divergence theorem*. It states that if $\vec{q}$ is C^0 and integrable, then

$$\int_{\Omega} \vec{\nabla} \cdot \vec{q} d\Omega = \oint_{\Gamma} \vec{q} \cdot \vec{n} d\Gamma \quad \text{or} \quad \int_{\Omega} \nabla^T \mathbf{q} d\Omega = \oint_{\Gamma} \mathbf{q}^T \mathbf{n} d\Gamma. \quad (12)$$

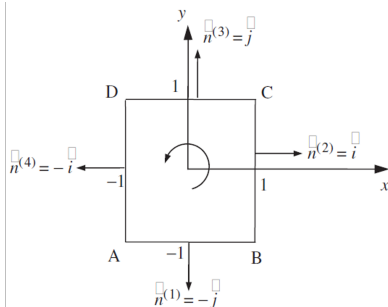
Green's formula, which is derived next, is the counterpart of integration by parts in one dimension. It states that

$$\int_{\Omega} w \vec{\nabla} \cdot \vec{q} d\Omega = \oint_{\Gamma} w \vec{q} \cdot \vec{n} d\Gamma - \int_{\Omega} \vec{\nabla} w \cdot \vec{q} d\Omega \quad \text{or} \quad \int_{\Omega} w \nabla^T \mathbf{q} d\Omega = \oint_{\Gamma} w \mathbf{q}^T \mathbf{n} d\Gamma - \int_{\Omega} (\nabla w)^T \mathbf{q} d\Omega. \quad (13)$$

Example 1

Given a rectangular domain as shown in the figure. Consider a scalar function $\theta = x^2 + 2y^2$. Let $\vec{q}$ be the gradient of θ defined as $\vec{q} = \vec{\nabla}\theta$. Contour lines are lines along which a function is constant.

- a) Find the normal to the contour line of θ passing through the point $x = y = 0.5$.
- b) Verify the divergence theorem for $\vec{q}$.



Domain used for illustration of divergence theorem.

Domain used for illustration of divergence theorem.

Example 1 (cont.)

The gradient vector $\vec{q}$ is given as

$$\vec{q} = \frac{\partial \theta}{\partial x} \vec{i} + \frac{\partial \theta}{\partial y} \vec{j} = 2x\vec{i} + 4y\vec{j}.$$

The gradient $\vec{q}$ is normal to the contour lines and its magnitude represents the slope of θ at any point. The gradient of θ at $x = y = 0.5$ is

$$\vec{q}(0.5, 0.5) = \vec{i} + 2\vec{j}.$$

At the point $x = y = 0.5$, the value of the scalar field θ is $\theta(0.5, 0.5) = 0.75$. The unit normal vector to the contour line $x^2 + 2y^2 - 0.75 = 0$ at the point $x = y = 0.5$ is obtained by dividing the vector $\vec{q}$ by its magnitude, which gives

$$\vec{n}(0.5, 0.5) = \frac{1}{\sqrt{5}} (\vec{i} + 2\vec{j}).$$

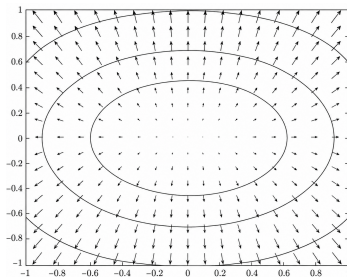


Figure 6.3 Contour lines of a function $\theta = x^2 + 2y^2$ and its gradient.

Example 1 (cont.)

We now verify the divergence theorem. The unit normal vectors at the four boundaries of the domain ABCD are shown in the figure. To verify the divergence theorem (12), we first evaluate the integrand on the LHS of (12):

$$\vec{\nabla} \cdot \vec{q} = \frac{\partial q_x}{\partial x} + \frac{\partial q_y}{\partial y} = 2 + 4 = 6.$$

Integrating the above over the problem domain gives

$$\int_{\Omega} \vec{\nabla} \cdot \vec{q} d\Omega = \int_{-1}^1 \left(\int_{-1}^1 6 dy \right) dx = 24.$$

Evaluating the boundary integral counterclockwise gives

$$\begin{aligned} \oint_{\Gamma} \vec{q} \cdot \vec{n} d\Gamma &= \int_{AB} (-4y) \underbrace{d\Gamma}_{dx} + \int_{BC} 2x \underbrace{d\Gamma}_{dy} + \int_{CD} 4y \underbrace{d\Gamma}_{-dx} + \int_{DA} (-2x) \underbrace{d\Gamma}_{-dy} \\ &= \int_{-1}^1 4 dx + \int_{-1}^1 2 dy + \int_{-1}^1 4 dx + \int_{-1}^1 2 dx = 24. \end{aligned}$$

Result

Thus, we have verified the divergence theorem for this example.

Example 2

Given a vector field $q_x = 3x^2y + y^3$, $q_y = 3x + y^3$ on the domain shown in the figure, verify the divergence theorem.

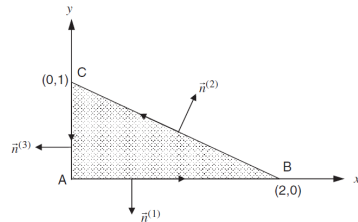


Figure 6.4 Triangular problem domain used for illustration of divergence theorem.

Triangular problem domain used for illustration of divergence theorem.

The integrand on the LHS of (12) is given as

$$\vec{\nabla} \cdot \vec{q} = \frac{\partial q_x}{\partial x} + \frac{\partial q_y}{\partial y} = 6xy + 3y^2.$$

Integrating the above yields

$$\int_{\Omega} \vec{\nabla} \cdot \vec{q} d\Omega = \int_0^2 \left[\int_0^{1-0.5x} (6xy + 3y^2) dy \right] dx = \int_0^2 \left[3x(1-0.5x)^2 + (1-0.5x)^3 \right] dx = 1.5.$$

The counterclockwise computed boundary integral on AB is

$$\int_{AB} \vec{q} \cdot \vec{n}^{(1)} d\Gamma = \int_{AB} q_x(-1) d\Gamma = \int_0^2 -3x dx = -6,$$

where $\vec{n}^{(1)} = -\vec{j}$, $d\Gamma = dx$ and $y = 0$ on AB. For the counterclockwise computed boundary integral on BC, note that the equation of the line BC is given by $y = 1 - 0.5x$ and $\vec{n}^{(2)} = \sqrt{5}/5 (\vec{i} + 2\vec{j})$, $d\Gamma = -\sqrt{5}/2 dx$ on BC. The boundary integral on BC is then given by

$$\int_{BC} \vec{q} \cdot \vec{n}^{(2)} d\Gamma = \int_{AB} (q_x \vec{i} + q_y \vec{j}) \frac{\sqrt{5}}{5} (\vec{i} + 2\vec{j}) d\Gamma = \int_2^0 -\frac{1}{2} \left[(3x^2 + y^3) + 2(3x + y^3) \right] dx = 7.75.$$

Finally, the counterclockwise boundary integral on CA is

$$\int_{CA} \vec{q} \cdot \vec{n}^{(3)} d\Gamma = \int_{AB} (q_x \vec{i} + q_y \vec{j})(-\vec{i}) d\Gamma = \int_1^0 y^3 dy = -0.25,$$

where $\vec{n}^{(3)} = -\vec{i}$, $d\Gamma = -dy$ and $x = 0$ on CA.

Adding the contributions of the three segments gives

$$\oint_{\Gamma} \vec{q} \cdot \vec{n} d\Gamma = \int_{AB} \vec{q} \cdot \vec{n} d\Gamma + \int_{BC} \vec{q} \cdot \vec{n} d\Gamma + \int_{CA} \vec{q} \cdot \vec{n} d\Gamma = -6 + 7.75 - 0.25 = 1.5,$$

which completes the demonstration that the divergence theorem holds for this case.

Result

Both the area integral and the boundary integral equal 1.5, confirming the divergence theorem.

Next: Weak Forms for Multidimensional Problems

Thanks for your attention!